

SUBHAM AMBASTHA



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EDUCATION

Expected in 07/2027

**Bachelor of Engineering (B.E.) -
Computer Science**

Dayananda Sagar Academy of
Technology & Management Bengaluru,
Karnataka

- CGPA: **8.98 / 10** (till 6th semester)
- 07/2022

**Higher Secondary Certificate (HSC) -
Science (PCM)**

Delhi Public School Bokaro Bokaro,
Jharkhand

- 89.33%** (Overall, 6 subjects)
- Ranked in Top **10%** of class

CERTIFICATIONS

- [Microsoft – Fundamentals of Computer Networking](#)
- [NPTEL - Foundations of Deep Learning](#)
- [The Complete Git Guide](#)
- [AI Fluency: Framework and Foundation](#)
- [Generative AI](#)
- [JavaScript](#)
- [Claude 101](#)
- [HTML5](#)
- [CSS3](#)

SUMMARY

Computer Science undergraduate with project experience in software development, backend systems, and machine learning. Skilled in Python and C++, with experience building applications and solving real-world problems.

SKILLS

Languages:	C++, Python, JavaScript, SQL
Core CS:	DSA, OOP, DBMS, Computer Networks
Backend:	FastAPI, REST APIs
Databases:	MongoDB, MySQL
ML:	TensorFlow, Scikit-Learn, XGBoost
Tools:	Docker, Git, Grafana, Prometheus

PROJECTS

F1 Strategy Engineer – ML-Powered F1 Pit Strategy Simulator

Tech Stack: Python, FastF1, XGBoost, Streamlit, Pandas, Plotly

- Developed an end-to-end Formula 1 strategy analysis platform using telemetry data from **33,569** racing laps across **23 Grand Prix** events and **27 drivers**.
- Engineered an **XGBoost**-based tyre degradation and lap-time prediction model achieving an **R² score of 0.973** on validation data.
- Designed a custom undercut simulation engine that identifies crossover laps and evaluates pit-stop strategies through cumulative undercut advantage analysis.

DevOps Observability Platform – Monitoring & Tracing System

Tech Stack: Docker, Prometheus, Grafana, ELK Stack, Jaeger, OpenTelemetry

- Built a production-style **observability platform** providing **centralized** monitoring, logging, distributed tracing, and automated alerting across multiple services.
- Implemented **OpenTelemetry**-based telemetry pipelines to collect application **metrics, traces, and logs**, enabling end-to-end visibility into system performance and reliability.
- Designed **unified Grafana** dashboards for tracking **latency, error rates, SLOs, deployments, and incident response metrics** to support operational monitoring and troubleshooting.

Vaak – Full-Stack Language Assistant

Tech Stack: FastAPI, MongoDB, JWT, REST APIs, Material UI, React

- Developed a full-stack language assistant supporting **multilingual translation, transliteration, dictionary lookup, and chat-based language assistance**.
- Designed and implemented secure authentication and user history management using **JWT-based authorization** and **RESTful APIs**.
- Integrated external translation and dictionary services to provide **real-time language detection**, translations, word definitions, and example usage.